

Education

King's College London <i>BEng Electronic Engineering, Predicted First-Class Honours</i>	London, UK Sept 2024 – Jun 2027
- Quantitative curriculum: Mathematical Modelling, Signals and Systems, Digital Logic, Electronic Principles, numerical modelling, and linear systems. - Research focus: financial time-series modelling, statistical signal extraction, backtesting methodology, and latency-aware systems.	

Selected Quantitative & Trading Experience

Jane Street FPGA Design Workshop <i>Funded Participant</i>	London, UK Feb 2026 – Apr 2026
- Selected for a six-week performance-critical systems program focused on deterministic logic design, parallel data paths, synchronization, timing constraints, and correctness under latency-sensitive conditions. - Implemented Verilog data-path logic, developing practical intuition for hardware-aware implementation, timing constraints, edge-case handling, and correctness under constrained design assumptions.	
Citadel Europe Data Open <i>Selected Participant & Datathon Competitor</i>	London, UK Feb 2026
- Selected to participate in a competitive data science datathon hosted by Citadel and Citadel Securities, collaborating in a team to analyse complex datasets under strict time constraints. - Leveraged Python, pandas, NumPy, and statistical modelling to clean data, construct features, test empirical hypotheses, and derive data-driven insights. - Presented methodology, model limitations, and conclusions to industry professionals, demonstrating fast research iteration and evidence-based analysis.	
Optiver Discover Tech-Driven Trading Event <i>Participant</i>	London, UK Nov 2025
- Attended a trading technology and quantitative finance event focused on market making, liquidity provision, and the integration of research in electronic markets. - Built Python simulations of market-making logic, bid-ask spread quoting, inventory exposure, and adverse-selection risk following the event.	

Research & Systems Projects

Global Macro Proxy Signal Research	Python, pandas, NumPy
- Designed a research framework testing trend, volatility-regime, and momentum signals across liquid macro proxies, including equity index ETFs, bond/rates proxies, FX pairs, and commodity ETFs. - Built an end-to-end pipeline covering price cleaning, rolling signal generation, volatility scaling, portfolio weighting, benchmark comparison, and transaction-cost modelling. - Evaluated signal robustness through walk-forward testing, drawdown decomposition, turnover analysis, annualised return, Sharpe ratio, maximum drawdown, and out-of-sample performance decay. - Focused on distinguishing persistent signal behaviour from noise by comparing performance across lookback windows, rebalance frequencies, and market regimes.	
Statistical Arbitrage & Cross-Sectional Factor Research	Python, statsmodels, Jupyter
- Developed an arbitrage workflow for liquid equities using correlation screening, spread construction, hedge-ratio estimation, z-score signal design, and mean-reversion diagnostics. - Built a single-factor research pipeline evaluating IC, Rank IC, ICIR, grouped return testing, and top-minus-bottom performance to assess factor predictiveness and stability. - Evaluated factor and pair-trading performance using Sharpe ratio, drawdown, turnover, transaction-cost sensitivity, and out-of-sample robustness checks. - Analysed common failure modes including look-ahead bias, unstable relationships, spread breakdown, overtrading, parameter instability, and transaction-cost erosion.	
FPGA Limit Order Book Matching Engine <i>Independent Systems Project, King's College London</i>	Verilog, GTKWave, Icarus Verilog Mar 2026 – Present
- Implemented and simulated core order-matching logic for add, cancel, and execute events, focusing on deterministic state transitions and hardware-aware design in Verilog. - Created test cases for order insertion, cancellation, execution, and queue-priority updates to verify logic correctness under diverse event sequences. - Used GTKWave to inspect timing behaviour and debug state-machine transitions for deterministic event processing.	

Leadership

KCL Snooker and Pool Club <i>President</i>	London, UK Mar 2026 – Present
Lead club operations, event planning, and stakeholder communication across student groups and external venues.	

Technical Skills

Quant & Trading	Market microstructure, trade analysis, statistical arbitrage, factor research, IC/Rank IC analysis, ICIR, transaction-cost modelling, walk-forward validation, robustness testing.
Mathematics	Time-series analysis, statistical reasoning, numerical modelling, linear systems, signal processing, optimisation-oriented modelling.
Programming Systems	Python, pandas, NumPy, matplotlib, scikit-learn, statsmodels, MATLAB, Git/GitHub, Jupyter Notebook. WSL2 Ubuntu, VS Code, Verilog, FPGA architecture, finite-state-machine design, deterministic event processing, GTKWave.
Languages	Mandarin Chinese native, English IELTS 7.5, Japanese basic.